Scientific Research Methodology
5. Identifying the research problem

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What is a research problem?

- the starting point of a research investigation
- a situation that is not fully understood
- the problem must be researchable

Researchable problems

Field	**Example of a researchable problem**	**Why it is researchable**
Education	How does immediate feedback affect pupils' performance?	Data can be collected and the hypothesis tested in class.
Psychology	Does meditation reduce physiologically measured stress levels?	Indicators such as cortisol or heart rate can be measured.
Physics	How does the acceleration of freely falling objects vary with mass?	Controlled experiments and measurements are possible.
Chemistry	How does the reaction rate between an acid and a metal depend on concentration?	Variables can be controlled and the experiment repeated.
Social sciences	Does employee productivity increase with remote work three days a week?	Data can be collected and analysed statistically.

Problems that are not researchable

Example of a non-researchable question	**Why it cannot be investigated scientifically**
What colour is the soul?	There is no method of observation or measurement.
Does the universe have an ultimate purpose?	It does not yield verifiable empirical data.
What is the most beautiful feeling?	This is a subjective judgement, dependent on personal values.
Is it better to be fair or happy?	This is a moral question, not an empirical one.
Is this theory true because "that is how I feel"?	The claim cannot be tested or falsified.

Sources for identifying a problem

- the specialist literature
- observation of real phenomena
- discussions with experts
- new situations or anomalies that current theory cannot explain

Characteristics of a well-formulated problem

- clear—unambiguous
- specific—defined within a particular context
- researchable—can be investigated using scientific methods

How do we formulate the problem?

- observe the situation: identify what is not known
- read the literature: identify a missing result, a "gap" in the field
- formulate the central research question

Example

Turn this statement into a research problem: **"Students use their phones too much."**

- Example: What is the relationship between time spent on the phone and academic performance?
- Methods: surveys, interviews, discussions
- Conclusions must be based strictly on the observations obtained in practice and on logical or statistical reasoning.

Example 2: the evolution of gravitational theory

Workflow

Problem $\longrightarrow$ Investigation $\longrightarrow$ Theory

The initial problem (Aristotle)

- Why do objects fall to Earth?
- Why do planets move across the sky?
- There was no unified explanation
- at that time, heavier objects were believed to fall "faster"

Copernicus, Kepler, Galileo

- observations + measurements + mathematical models
- Copernicus: the Sun at the centre
- Kepler: the laws of planetary motion: elliptical orbits, three mathematical laws verifiable through observation. These contradict the theory of circular orbits
- Galileo: experiments on falling objects (the Tower of Pisa)—objects are observed to fall with the same acceleration, regardless of mass!

A new theory—Newton

- a universal force exists between any two bodies
- Mathematical formulation: $F = gm_1m_2/r^2$
- It explains both falling objects and planetary motion.

The theory continues to evolve

- A new phenomenon: gravity affects light
- General relativity: gravity is not a force but the curvature of spacetime caused by mass and energy
- Gravity = the deformation of spacetime

Example 3: the evolution of the atomic model

Workflow

Problem $\longrightarrow$ Investigation $\longrightarrow$ Theory

The initial problem:

- What is matter made of? Are there "basic" elements?
- Why do chemical reactions occur in fixed proportions?

Dalton: the indivisible atom

Atoms are described as small units of matter: spherical, indivisible, and solid.

- measurements of the reactants and products of chemical reactions
- constant proportions are observed
- Theory: atoms are solid, indivisible spheres

Thomson: the discovery of the electron

The electron is discovered.

- investigation: experiments with cathode-ray tubes
- negatively charged particles (electrons) are discovered
- Theory: the atom is a positively charged sphere surrounded by electrons

Later stages of the theory

Rutherford: the atomic nucleus

The atom has a small, positively charged nucleus, with negatively charged electrons orbiting it like planets around the Sun.

- experiments using gold foil demonstrated the existence of subatomic particles
- Theory: the atom has a small, positively charged nucleus

Bohr: a model with energy-defined orbits

Electrons orbit the nucleus in fixed orbits according to their energy level.

- the emission spectrum of hydrogen
- Theory: transitions between levels emit or absorb energy

An example from the social sciences

Learning theory: Problem $\longrightarrow$ Investigation $\longrightarrow$ New theory

Stage / Author	Problem	Investigation	Resulting new theory
Jean Piaget (1920–1970)	How does intelligence develop? Do children learn simply by accumulating information?	Systematic observations, interviews with children, simple experiments (conservation of quantity, etc.)	Theory of cognitive development in **stages**. Development is determined by internal maturation + experience.
Lev Vygotsky (1970–1980)	Why does social context influence learning if development is universal?	Observing child–adult interaction, the role of language and guided learning	**Social constructivism**. The concept of the *Zone of Proximal Development (ZPD)*.
Modern theories (1990–present)	Can we measure and personalise learning in real time?	Neuroimaging (fMRI), computational models, e-learning data	Learning as a **neurocognitive + social + technological** process. Neuroplasticity + personalisation.

Conclusion: the scientific research process

Workflow

Problem $\longrightarrow$ Investigation $\longrightarrow$ Theory

- scientific theories are refined as new data emerge
- every unexplained phenomenon challenges the existing theory and motivates the search for a new one
- the first step: choosing a problem. Look for examples!